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Bachelor of Science in Information Technology**



**Designing a Smart City Information System to Enhance Public
Service Delivery and Citizen Engagement in Olongapo City**

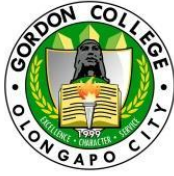
A Capstone Project Presented By:

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CHAPTER 1



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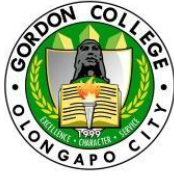


THE PROBLEM AND ITS BACKGROUND

This chapter provides an overview of the study by presenting the Background of the study, the theoretical and conceptual frameworks, and the statement of the problem. It further outlines the scope and limitations, describes the conceptual framework, discusses the significance of the study, and defines the key terms used.

1.1 Background of the Study

In recent years, several cities around the world have utilized technology to enhance the delivery of public services and improve their interactions or engagement with citizens. The idea is often called "Smart City"; a smart city is a city that combines information and communication technologies, social infrastructure (human and social impact), and public institutions to dynamize its economic, social, environmental, and cultural development (Suwar & Kozlowski, 2021). Through this approach, people could participate more actively in decision-making and community activities. Additionally, while smart cities rely heavily on information and communication technologies (ICT) to deliver services, the effectiveness of these systems depends on active citizen engagement and the local government's ability to respond efficiently to community needs.

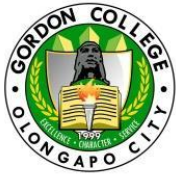


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These principles are being adopted by smart city initiatives in several countries worldwide. For instance, Singapore uses a mobile app called OneService, which allows the public to submit feedback on municipal issues such as cleanliness and infrastructure through a single platform (Ministry of National Development Singapore, 2023). Similarly, Seoul, South Korea, operates a "Smart Complaint Centre" and an Integrated Citizens Complaint and Comprehensive Advice Service (CCPIS), where residents can send complaints or suggestions to city officials via mobile and online channels (Seoul Solution, 2023). Additionally, Barcelona, Spain, has implemented an application called the Sentilo platform, along with sensor-based systems, to manage garbage collection, traffic, and real-time street lighting, thereby improving efficiency and transparency in city services (City of Barcelona, 2023). These demonstrate how various governments worldwide are utilizing technology to deliver faster, more transparent, and citizen-centric public services.

In the Philippines, local governments have increasingly taken on smart city projects to improve public service delivery and boost citizen engagement. For example, Makati City created a mobile app that allows residents to report emergencies, give feedback, and communicate local issues directly to city officials (GMA News, 2019). These digital tools show



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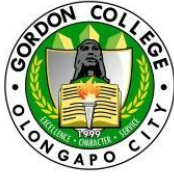


how technology can simplify government processes, increase responsiveness, and encourage better interaction between citizens and local leaders. By enabling real-time communication and more effective service management, these systems help make governance more transparent, accountable, and focused on citizens. Understanding how these initiatives are implemented and their impact is crucial for assessing the Smart City Information System (SCIS) in Olongapo City, especially regarding its role in supporting citizen engagement, service efficiency, and overall satisfaction with local government services.

Examining both international and local experiences shows that smart city projects can greatly enhance public services and motivate citizens to engage more in their communities. However, the success of these systems depends not only on the technology but also on how well people embrace and use it, as well as how responsive local governments are to the needs of citizens. This underscores the need to evaluate the Smart City Information System in Olongapo City. Understanding its effectiveness can lead to better service delivery, promote genuine citizen involvement, and inform the development of future digital projects that truly benefit the community.

1.2 Theoretical and Conceptual Framework

1.2.1 Theoretical Framework



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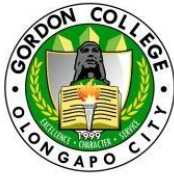


This section of the study revolves around theories that explain how technology and user engagement could improve the effectiveness of public service delivery and foster citizen participation through a Smart City Information System (SCIS) in Olongapo City.

Socio-Technical Systems Theory (STS).

Socio-Technical Systems (STS) highlights that the effectiveness of any system depends on the interaction and joint optimization of two interrelated subsystems, namely, technical and social. The technical subsystem includes tools, processes, and technologies, while the social subsystem, on the other hand, encompasses people, organizational structures, roles, and interactions (Trist & Bamforth, 1951).

The theory is used as a basis for evaluating the Smart City Information System (SCIS) of Olongapo City, as the theory provides a comprehensive framework where the technical subsystem consists of the SCIS infrastructure, software, databases, and communication networks, and the social system, on the other hand, involves citizens and city employees who interact with the system. Applying STS theory allows the study to assess delivery and citizen engagement, emphasizing that the system's success depends not only on technological efficiency but also on user collaboration, adoption, and organizational support.



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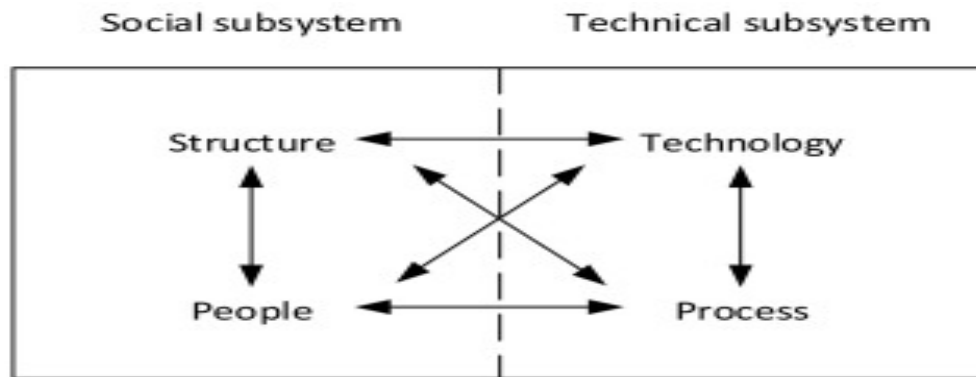
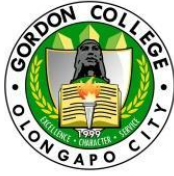


Figure 1. Socio-Technical System Theory (STS)

Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) was developed by Davis (1989). The model provides a framework for understanding how users adopt and utilize new technologies. The core structure of the Technology Acceptance Model is the Perceived Usefulness (U), which refers to the degree to which users believe that the system enhances their performance, and Perceived Ease of Use (E), the degree to which users find the system effortless to use in which could influence the Behavioral Intention (BI) and, eventually, the Actual System Use.

The model is applied to evaluate the acceptance and use of the Smart City Information System (SCIS) in Olongapo City. Particularly, the model could aid the researchers in determining whether citizens and city employees perceive the SCIS as useful in providing public service delivery and



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enhancing its efficiency and accessibility. By examining these factors, TAM enables the study to assess the degree of user engagement with the SCIS and its impact on public service delivery and citizen participation. In addition, understanding these user perceptions could provide valuable insights for improving the system design, training, and communication strategies, ensuring that the SCIS effectively meets the needs of both citizens and government personnel.

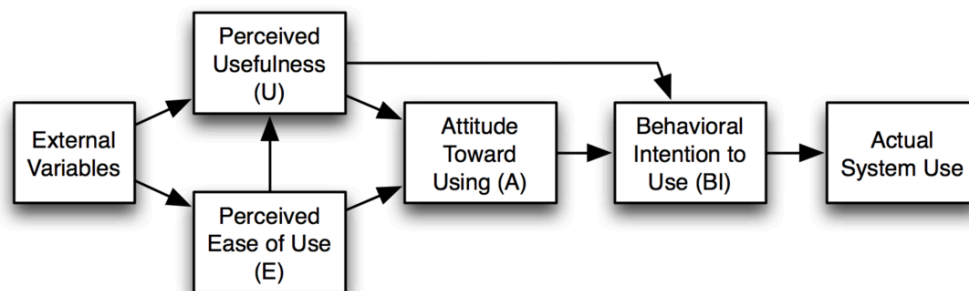
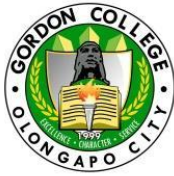


Figure 2. Technology Acceptance Model (TAM)

Unified Theory of Acceptance and Use of Technology (UTAUT)

Unified Theory of Acceptance and Use of Technology (UTAUT) extends the understanding of technology acceptance by integrating elements from several previous models, including the Technology Acceptance Model (TAM), to provide a covering framework for predicting and explaining user behavior

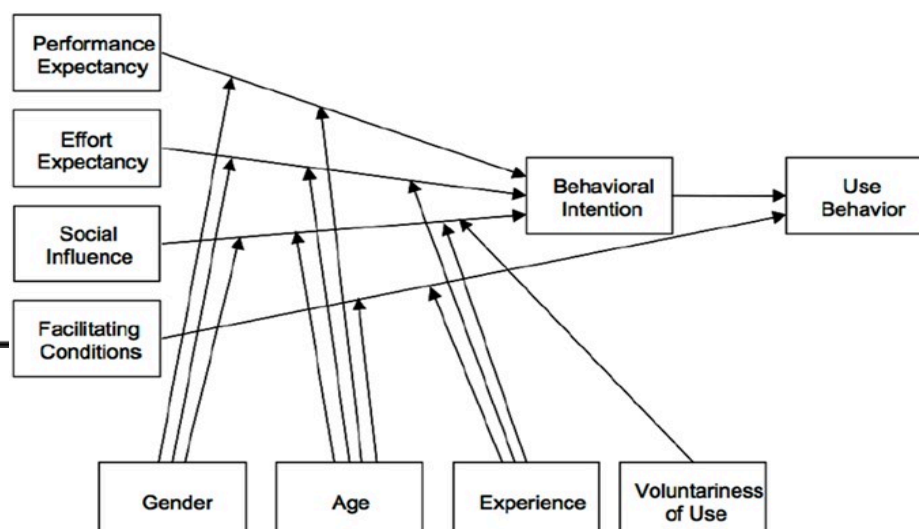


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(Vankatesh et al., 2003). UTAUT outlines four main factors, namely, Performance Expectancy, which reflects the perceived benefits and improvement in performance using the system; Effort Expectancy, which captures how users perceive the ease of use of the system; and Social Influence, which measures the effect of others' opinions on the user's decision to adopt the system. Lastly, Facilitating Conditions, which refers to the availability of technical and organizational support for system use.

In Olongapo City, UTAUT provides a framework for evaluating how social factors, organizational support, and perceived benefits influence both citizens' and employees' as well as their actual usage. With that being said, UTAUT offers a more comprehensive understanding of technology adoption, involving not only individual perceptions of usefulness and ease of use but also the social and institutional conditions that affect the successful implementation and effectiveness of the SCIS in enhancing public service delivery and citizen engagement.



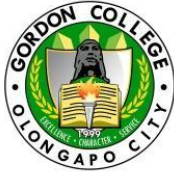


Figure 3. Unified Theory of Acceptance and Use of Technology (UTAUT)

1.2.2 Conceptual Framework

The conceptual framework of this study is anchored on the integration of Socio-Technical Systems Theory, the Unified Theory of Acceptance and Use of Technology (UTAUT), and the Technology Acceptance Model (TAM). It illustrates the dynamic relationship between the Smart City Information System (SCIS) implementation in Olongapo City and its outcomes in enhancing public service delivery and citizen engagement.

The framework posits that the effectiveness of SCIS is determined by the interplay between technical subsystem factors (system usability, accessibility, trust and security, institutional readiness) and social subsystem factors (performance expectancy, effort expectancy, social influence, facilitating conditions), which collectively influence user acceptance and system utilization.

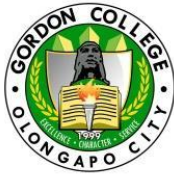


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These factors subsequently lead to two primary outcomes: (1) enhanced public service delivery, characterized by improved service efficiency, transparency and accountability, service quality, timeliness, cost-effectiveness, and reliability; and (2) enhanced citizen engagement, manifested through accessible engagement channels, government responsiveness, active participation and involvement, trust in government, effective feedback mechanisms, and community collaboration.

The framework provides a comprehensive structure for assessing how the collective contribution of technological implementation and user adoption contributes to achieving smarter governance and more engaged citizenship in Olongapo City.



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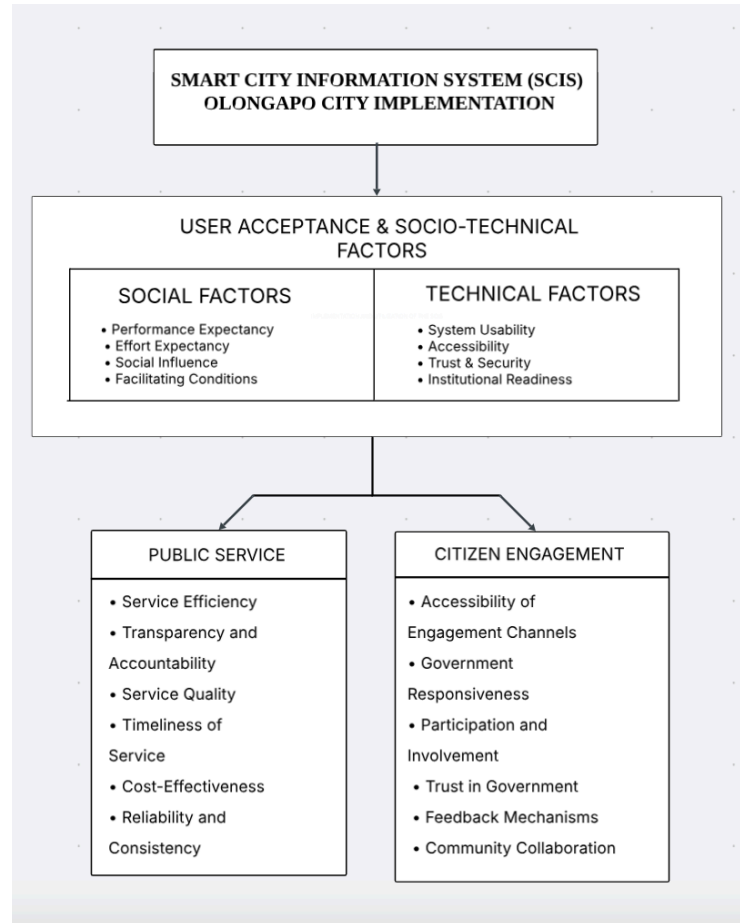


Figure 4. Conceptual Model of Olongapo City Smart City Information System.

1.3 Statement of the Problem

This study aims to assess the effectiveness of a Smart City Information System in enhancing citizen engagement and public service



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delivery in Olongapo City. It examines how the system could improve the services provided by the Olongapo City Government.

The study specifically seeks to answer these questions:

1. What is the demographic profile of the respondents in terms of:
 - 1.1 Age;
 - 1.2 Sex;
 - 1.3 Access to internet/technology;
2. What is the current level of citizen engagement through existing platforms in terms of:
 - 2.1 Accessibility of engagement channels;
 - 2.2 Local government responsiveness;
 - 2.3 Participation and involvement;
3. How effective is the Smart City Information System in addressing challenges in governance in terms of:
 - 3.1 Transparency;
 - 3.2 Service efficiency;
 - 3.3 Citizen engagement;
4. What factors affect the acceptance and utilization of a Smart City System in terms of:



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- 4.1 Usability;
 - 4.2 Accessibility;
 - 4.3 Trust and security;
 - 4.4 Institutional readiness;
5. What are the challenges and limitations that may hinder the full adoption of the Smart City Information System, particularly in terms of:
- 5.1 Digital literacy and technology gaps;
 - 5.2 Data security and privacy issues;
 - 5.3 Resistance to adaptation from citizens or government employees;

1.4 Significance of the study

The researcher anticipates that the findings of this study will provide valuable insights not only for students but also for other stakeholders. Specifically, by assessing the effectiveness of a Smart City Information System in enhancing public service delivery and citizen engagement in Olongapo City, this research is expected to benefit the following groups:



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Olongapo City Citizens. This study could benefit the citizens of Olongapo City by identifying how effectively the system could improve access to public services and increase opportunities for civic participation. This, in result, could lead to a more engaged and informed community, fostering a stronger sense of belonging and responsibility among residents. By leveraging technology and innovative approaches, the city can enhance transparency and accountability, ultimately empowering its citizens.

Local Government Officials. This study could benefit local government officials by identifying ways to improve access to public services and increase opportunities for civic participation. This information could empower officials to implement more effective policies and engage the community more deeply. By understanding the needs and preferences of residents, they can foster a more inclusive environment that encourages collaboration and feedback.

System Developers and Implementers. This study could provide valuable insights for system developers and implementers by offering data on user satisfaction and system functionality, which may inform upgrades or the design of similar platforms in other localities. Such feedback can guide developers in making evidence-based decisions, ensuring that future



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iterations of the system are more user-friendly and effective. Furthermore, understanding user preferences may inspire innovative features that enhance overall engagement, usability, and the quality of service delivery.

Students and Educational Institutions. This research could become a helpful resource for students in information technology, public administration, urban planning, and similar fields. By presenting a real example of smart city implementation, students can learn about the connection between technology, governance, and citizen involvement, which enhances their academic and professional growth.

Policy Makers and Community Organizations. This study could offer useful guidance for policymakers, NGOs, and community organizations. It highlights effective practices, challenges, and strategies for adopting technology in local governance. These insights can support larger efforts to improve digital governance and foster public services that focus on citizens.

Future Researchers. This study is expected to serve as a valuable reference for scholars and practitioners who wish to explore similar topics. By providing empirical data and analysis, it could guide future investigations on smart city information systems and related areas.



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In summary, the results of this study are envisioned to contribute not only to the continuous improvement of Olongapo City's Smart City Information System but also to the broader understanding of how technology can strengthen governance, enhance service delivery, and deepen citizen engagement in other communities.

1.5 Scope and Limitation of the Study

This study focuses on assessing the effectiveness of the Smart City Information System (SCIS) of Olongapo City in enhancing public service delivery and citizen engagement. The research involves residents of Olongapo City as the primary respondents, with additional insights from local government officials and stakeholders involved in the system's management. The SCIS will be evaluated in terms of three key dimensions: service efficiency, transparency, and citizen engagement. Data collection will be conducted during the academic year 2025, focusing on how the system supports faster and more reliable service delivery, provides accessible and transparent government information, and promotes active citizen participation in governance.



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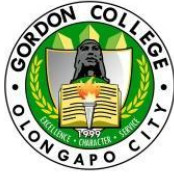


However, the study's results are limited to the context of Olongapo City and may not be fully generalizable to other areas with different demographics or technological infrastructures. The research also depends on the willingness and availability of citizens to participate, which may exclude those with low digital literacy or limited internet access. Since the study relies on self-reported data, responses may reflect personal bias or incomplete recollections. Furthermore, the assessment is confined to the features currently available in the SCIS and does not consider future updates or system improvements. External challenges, such as poor connectivity, socio-economic disparities, and resistance to technology adoption, may also impact the system's perceived effectiveness.

1.6 Definition of Terms

The following key terms are defined conceptually and/or operationally to ensure clarity and consistency of meaning throughout the study:

Citizen Engagement. Involvement of the residents in government programs, feedback mechanisms, and various decision-making processes, including interaction with the Smart City Information System.



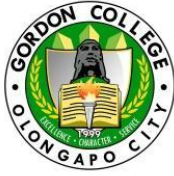
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Effectiveness. Effectiveness can be defined as the aptitude to fulfill the desired purpose (Gulluscio, 2014). In the context of the study, it refers to how effectively the program achieves its intended objectives.

Public Service Delivery. According to Afeez Kolawole Shittu (2022), service delivery refers to “the extent to which services provided by public sector organizations meet or exceed the expectations of beneficiary groups, which include the general public.” In this study, the term pertains to the delivery of government services through Olongapo City’s Smart City Information System, focusing on how effectively it meets citizens’ needs and expectations.

SCIS (Smart City Information System). Conceptually, smart cities are defined by the British Standards Institute (2014) as “the effective integration of physical, digital, and human systems in the built environment to deliver a sustainable, prosperous, and inclusive future for its citizens.” In the context of this study, the Smart City Information System refers to the integrated digital platform adopted by Olongapo City that embodies this definition by consolidating government services, information, and communication channels to improve efficiency and responsiveness.



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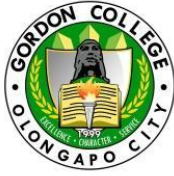
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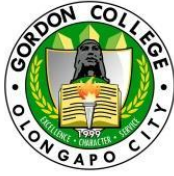
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